Numerical Methods and Optimization

28/4/2024 – 3^{rd} Year

ENSIA — Spring 2024

Tutorial 8

- Classical methods of optimization -

Exercise 1:

Find the extremum of:

$$f(x, y, z) = x^2 - 2xy + y^2 + 5z^2$$

subject to the constraint:

$$x + y + 2z = 10$$

Exercise 2:

A company manufactures two types of machines: x and y. The joint cost function is given by:

$$f(x, y) = x^2 + y^2 - 10x - 12y + 151$$

(1) Find the number of machines of each type that the company must manufacture to minimize its cost.

(2) Using second-order conditions, demonstrate that it is indeed a minimum.

Exercise 3:

A monopolistic firm produces a certain good. The total cost of its production is represented by the function:

$$C = 2I^2 + 5q^2 - 20q + 400$$

Where q represents the units produced and I an index of product quality (for example, the degree of product grading). The price that the firm can demand in the market depends on the quality index: $p = 100 - 3q + 4I$

(1) Calculate the total revenue of the firm.

(2) Find q and I such that they maximize the firm's profit.

(3) Verify the second-order conditions.

Exercise 4:

You are flying charter for a trip to the United States. In your suitcase, you are allowed to carry $24kg$. Your toiletries weigh $4kg$, leaving you with $20kg$ to use optimally. You would like to bring three types of goods: *blue jeans* denoted **X**, *T-shirts* denoted **Y**, and *sweaters* denoted **Z**. The respective unit weights are (in kg):

$$p_X = 2 \qquad p_Y = 0.5 \qquad p_Z = 1$$

The utility these clothes provide you during your trip is measured by the function:

$$U = X^2YZ$$

(1) Choose X, Y, and Z to maximize utility.

(2) Having chosen your optimal assortment, you realize that you cannot close your suitcase. The total available volume (after storing your toiletries) is $40dm^3$. The respective unit volumes of the three clothing items (in dm^3) are:

$$V_X = 4 \qquad V_Y = \frac{2}{3} \qquad V_Z = 3$$

Find the optimal combination under the **dual constraint** of weight and volume.

Exercise 5:

A firm produces a certain good that it sells in the market at the unit price of 8$. The quantity produced (and sold) is given by:

$$Q = 8K^{\frac{3}{8}}L^{\frac{1}{8}}$$

where K represents the units of capital employed and L the units of labor employed.Therefore, the firm's revenue amounts to

$$8Q = 64K^{\frac{3}{8}}L^{\frac{1}{8}}$$

The production cost of this firm is given by: $C = 12K + 4L$.

(1) Find the quantities of K and L that the firm must employ to maximize its profit.

(2) The firm realizes that the profit-maximization policy does not ensure a sufficient market share. Consequently, it decided to adopt the policy of maximizing the quantity sold, subject to a minimal profit equal to 48$.

i- Explicitly formulate the maximization problem under constraint.

ii- Show that in the optimal solution, the optimal combination of factors (i.e., their proportions) remains the same as in the profit-maximization problem.

iii- Provide the complete solution for K, L, and Q.